

Lab 1 - Hyperelasticity Modelling

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I. OBJECTIVES AND EXPERIMENTAL SETUP

The purpose of this experiment was to test the stress on two polymers, Dragonflex and Ecoflex, by adding weight until either the polymer snapped or we collected 10 data points.

To start, we created two dog bone-shaped molds using cardboard and printed templates. We layered cardboard for structure, taped the templates on top, and laminated everything to prevent the polymer from sticking. After cutting out the molds, we assembled them with hot glue, ensuring the laminated side faced inward. (Images in Appendix)

We then prepared the polymer mixtures by adding 12.5 ml each of Part A and Part B for both Dragonflex and Ecoflex. After pouring the mixtures into the molds halfway, we added small fabric pieces at each end as anchor points for attaching the weights. We filled the molds, let them cure in the oven for 20 minutes, and then prepared for testing.

For the test, we secured one end of the dog bone to the table with tape and attached a 2L empty water bottle to the other end. We added 100 ml of water at a time and measured how much the polymer stretched. This was done in 10 increments until the polymer either snapped or we reached the limit.

The results showed that Dragonflex held its shape better than Ecoflex, which stretched more easily under the same load. We successfully collected 10 data points for both polymers.

II. RESULTS & ANALYSIS

Young's modulus is a stress-to-strain ratio in elastic materials. It only applies to small deformations, and several other properties can affect its usability (temperature, viscoelasticity, etc). When applying $<0.2\%$ deformation, the modulus is considered accurate. In terms of Dragon Skin 10 and Ecoflex 00-30, we can learn about the uniaxial stress behavior, however, it only applies to these small deformations (likely only our first few results with little strain if at all). In the Figure 1 below, the deformations of each polymer are shown for each 100 mL of water added to the bottle.

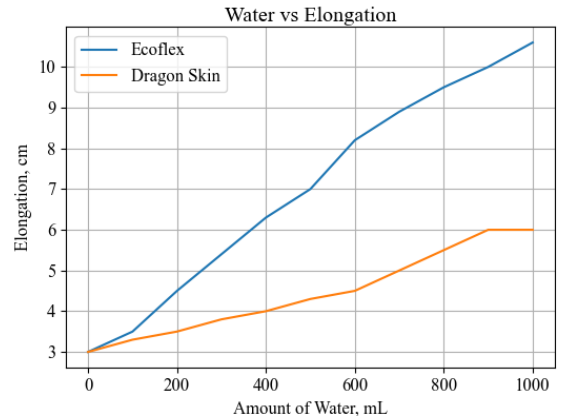


Figure 1: Water vs Elongation

Our results align with existing hyperelasticity models, which indicate that Dragon Skin exhibits less stretch compared to Ecoflex. In our experiment, Dragon Skin experienced less elongation under the same load than Ecoflex, but it still stretched rather than fully maintaining its shape.

Our data seemingly matches the database experimental data available on the SRMD relatively well. Despite minor differences in the curves (likely due to experimental error on our part) and 'jumpy' data, they are still comparable in both values and curves. Additionally, our coefficients seem to be somewhat similar to each other. This is seen in the almost identical overlap of the Poly and Ogden models. The two plots are shown below in Figure 2 and Figure 3.

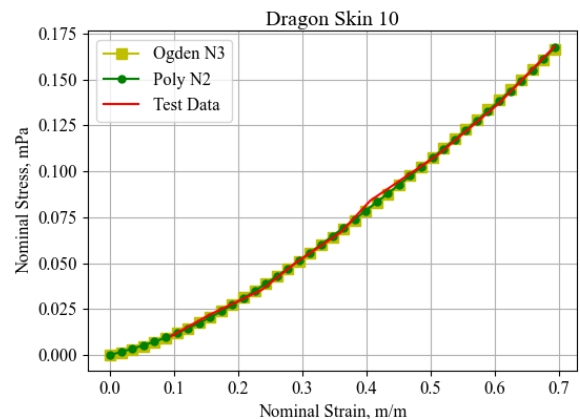


Figure 2: DragonSkin Hyperelastic Model

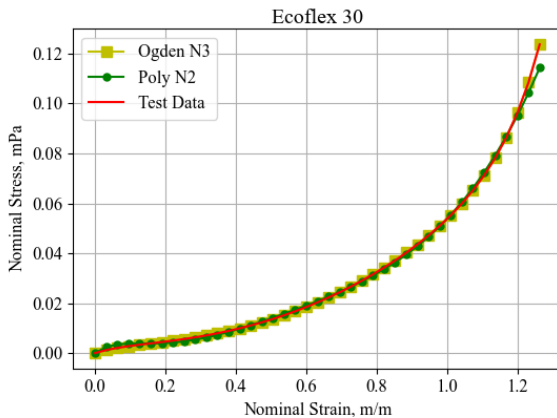


Figure 3: EcoFlex Hyperelastic Model

The two samples that were tested did not fail. This means that a failure strain could not be calculated for either the Dragonflex or Ecoflex materials. However, if one of the dog-bones did fail it could be expected that the value of the Ecoflex failure strain to be larger than the Dragonflex. This is due to the Ecoflex having more elongation in tension than the Dragonflex.

We didn't directly compare our data with other groups, however we would expect their results to be similar to ours. Variations between our data and the data from other groups would most likely occur due to differences in constructing and filling the mold. The most significant difference we did observe from this is that some groups had their samples fail, while ours did not.

If the experiment was run again, it would be expected that the material properties vary from our original batch. Since the A:B ratio was estimated, creating a second batch would almost certainly result in the cured material being slightly different. However, how the material properties change, or whether it would noticeably change at all, would require some testing.

III. ASSUMPTIONS AND APPROXIMATIONS

One assumption that was made during the experiment is that the samples were properly and uniformly produced. Due to human error, there were probably bubbles in the materials, as well as some deviation from the 1:1 ratio. Additionally, we assumed the sample was the desired height, width, and thickness. All of these human errors could contribute to deviations between the experimental data and models of existing data.

IV. CONCLUSION

From the reading, *Toward a Common Framework and Database of Materials for Soft Robotics*, there are many

challenges and aspects related to characterizing the behavior of soft materials for soft robotics. One major difficulty is the ability to control morphology and stiffness in a reliable, repeatable manner, as soft materials are highly deformable and susceptible to environmental factors. Soft materials also face challenges to be accurate, reliable, and have repeatable testing. One must account for additional strain rates, and there is a need for more experimental data for proper calibration and accurate characterization. Finally, soft materials often have an underestimate of induced strain and an overestimate of stiffness. The aspects related to characterizing the behavior of soft materials includes methods and guidelines of testing, the strain rate, the Principal Cauchy stress, and FEA software for true data, to name a few.

APPENDIX



Image 1: Empty Mold

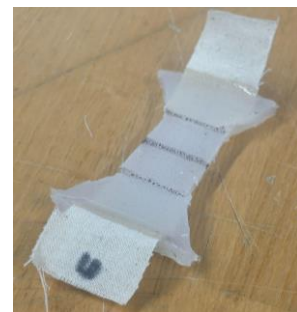


Image 2: Final Sample



Image 3: Filling with Water